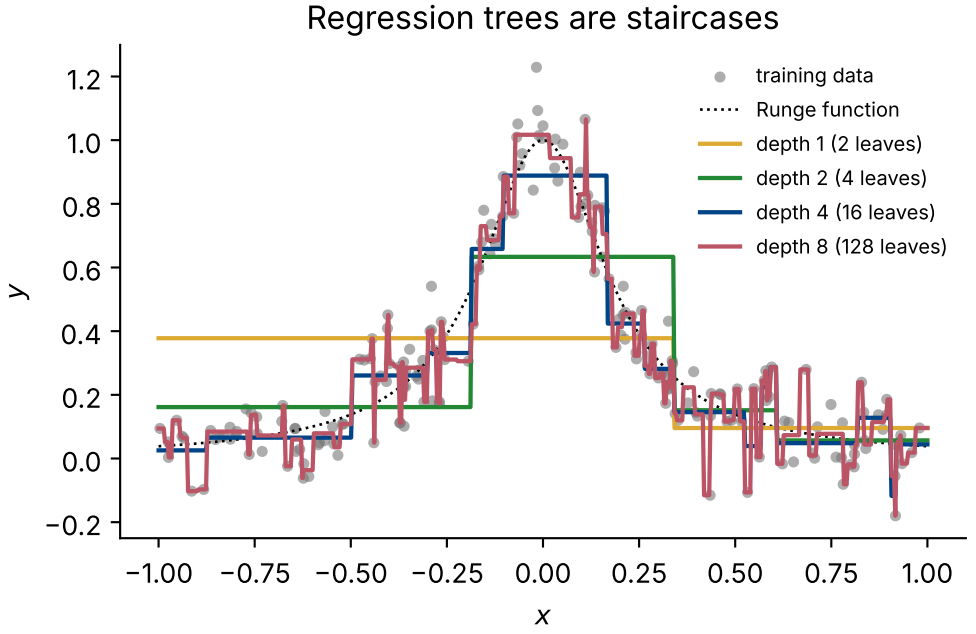
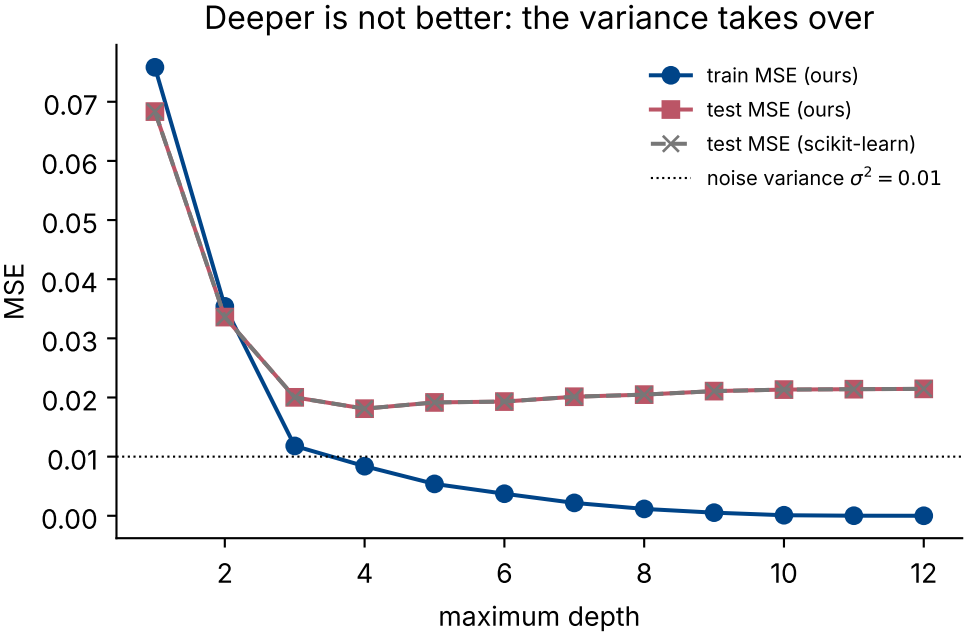

Regression trees are staircases
training data
Runge function
depth 1 (2 leaves)
depth 2 (4 leaves)
depth 4 (16 leaves)
depth 8 (128 leaves)
y
x
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


Deeper is not better: the variance takes over
train MSE (ours)
test MSE (ours)
test MSE (scikit-learn)
noise variance $\sigma^2 = 0.01$
MSE
maximum depth
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
2
4
6
8
10
12